INDEX

S.NO:	CHAPTER	PAGE NUMBER
1	Reproduction in Organisms	1
2	Sexual Reproduction in Flowering Plants	2
3	Human Reproduction	3
4	Reproductive Health	4
5	Principles of Inheritance and Variation	5
6	Molecular basis of Inheritance	6
7	Evolution	7
8	Human Health and Disease	8
9	Strategies for enhancement in food production	9
10	Microbes in Human Welfare	10
11	Biotechnology: Principles and Processes	11
12	Biotechnology and its Applications	12
13	Organisms and Populations	13
14	Ecosystem	14
15	Environmental Issues	15
16	Biodiversity and Conservation	16

Practice perfects!

CLASS: XII

SUBJECT: BIOLOGY

CHAPTER NUMBER: 1

CHAPTER: REPRODUCTION IN ORGANISMS

ASSERTION AND REASONING TYPE OF QUESTIONS WITH ANSWERS

These questions consist of two statements each, printed as Assertion and Reason. While answering these questions you are required to choose any one of the following four responses.

A . If both Assertion and Reason are true and the Reason is correct explanation of the Assertion.

B. If both Assertion and Reason are true but the Reason is not a correct explanation of the Assertion.

C. If Assertion is true but the Reason is false.

D. If both Assertion and Reason are false

****

1. **Assertion:** Zygote is the only cell that gives a vital link between two generations of an organism.

   Reason: The two gametes fuse to form a single zygote.

2. **Assertion:** Viviparous animals give better protection to their offspring.

   Reason: They lay their eggs in safer places in the environment.

3. **Assertion:** An ovule is transformed into a fruit after fertilization.

   Reason: The fruits are formed only from ovules.

4. **Assertion:** In angiosperms, the ovule develops into a seed after fertilization.

   Reason: Fertilization is not essential for the development of fruit.

5. **Assertion:** A plant biotype can be retained and multiplied indefinitely without any change or variation by the method of asexual reproduction

   Reason: Asexual reproduction does not involve meiosis and syngamy.

ANSWERS:

1. B 2. C 3. D 4. B 5. A

CLASS: XII

SUBJECT: BIOLOGY

CHAPTER NUMBER: 2

CHAPTER: SEXUAL REPRODUCTION IN FLOWERING PLANTS

ASSERTION AND REASONING TYPE OF QUESTIONS WITH ANSWERS

These questions consist of two statements each, printed as Assertion and Reason. While answering these questions you are required to choose any one of the following four responses.

A . If both Assertion and Reason are true and the Reason is correct explanation of the Assertion.

B. If both Assertion and Reason are true but the Reason is not a correct explanation of the Assertion.

C. If Assertion is true but the Reason is false.

D. If both Assertion and Reason are false

****

1. **Assertion:** Red colour of flowers attracts butterflies and wasps, but not bees.

   Reason: Bees are colour-blind to red.

2. **Assertion:** Cross pollination results in healthy and stronger offspring.

   Reason: Due to phenomenon of hybrid vigour.

3. **Assertion:** The endosperm of angiosperms is generally triploid (3n)

   Reason: It develops from primary endosperm nucleus formed by fusion of haploid male gametes and diploid secondary nucleus.

4. **Assertion:** Self-pollination occurs in Pteridophytes and monocots.

   Reason: Cross-pollination occurs in gymnosperms and dicots

5. **Assertion:** Insects visit flowers to gather honey.

   Reason: Attraction of flowers prevents the insects from damaging other parts of the plant.

ANSWERS:

1. A 2. A 3. A 4. D 5. D

CLASS: XII

SUBJECT: BIOLOGY

CHAPTER NUMBER: 3

CHAPTER: HUMAN REPRODUCTION

ASSERTION AND REASONING TYPE OF QUESTIONS WITH ANSWERS

These questions consist of two statements each, printed as Assertion and Reason. While answering these questions you are required to choose any one of the following four responses.

A . If both Assertion and Reason are true and the Reason is correct explanation of the Assertion.

B. If both Assertion and Reason are true but the Reason is not a correct explanation of the Assertion.

C. If Assertion is true but the Reason is false.

D. If both Assertion and Reason are false

****

1. Assertion: In humans, the gamete contributed by the male determines whether the child produced will be male or female.

Reason: Sex in humans is a polygenic trait depending upon a cumulative effect of some genes on X-chromosome and some on Y-chromosome

2. Assertion: In the testes, spermatogenesis occurs in the seminiferous tubules and testosterone secretion takes place in the interstitial cells.

Reason: Testosterone brings about growth and maturation of secondary sex organs and also development of accessory sex characters.

3. Assertion: In a Graafian follicle, the primary oocyte and the follicle cells may be regarded sibling cells.

Reason: Both arise from the same parent cell, the oogonium, by mitotic divisions.

4. Assertion: Ovum retains most of the contents of the primary oocyte and is much larger than a spermatozoon.

Reason: Ovum needs energy to go about in search of a spermatozoon for fertilization.

5. Assertion: A sperm sticks to an egg for fertilization.

Reason: Interaction of surface receptors, fertilizin on the egg and antifertilizen on the sperm head, makes them adhere together.

ANSWERS:

1. C 2. B 3. A 4. C 5. A

CLASS: XII

SUBJECT: BIOLOGY

CHAPTER NUMBER: 4

CHAPTER: REPRODUCTIVE HEALTH

ASSERTION AND REASONING TYPE OF QUESTIONS WITH ANSWERS

These questions consist of two statements each, printed as Assertion and Reason. While answering these questions you are required to choose any one of the following four responses.

A . If both Assertion and Reason are true and the Reason is correct explanation of the Assertion.

B. If both Assertion and Reason are true but the Reason is not a correct explanation of the Assertion.

C. If Assertion is true but the Reason is false.

D. If both Assertion and Reason are false

****

1. **Assertion:** Use of condom is a safeguard against AIDS and sexual diseases besides checking pregnancy.

   Reason: Certain contraceptives are planted under the skin of the upper arm to prevent pregnancy.

2. **Assertion:** Amniocentesis is often misused.

   Reason: Amniocentesis is meant for determining the genetic disorders in the foetus, but is being used to determine the sex of the foetus so that female foetus may be aborted.

3. **Assertion:** Human population now doubles every 35 years as against 200 years in 1600-1800.

   Reason: Rapid increase is due to better health facilities and food resources.

4. **Assertion:** Zero population growth should be achieved as early as possible to control human population.

   Reason: This requires not two children per couple but a little more.

5. **Assertion:** Mother should not be blamed for the birth of girls in the family.

   Reason: Father is responsible for the sex of the child.

ANSWERS:

1. B 2. A 3. A 4. A 5. B

CLASS: XII

SUBJECT: BIOLOGY

CHAPTER NUMBER: 5

CHAPTER: PRINCIPLES OF INHERITANCE AND VARIATION

ASSERTION AND REASONING TYPE OF QUESTIONS WITH ANSWERS

These questions consist of two statements each, printed as Assertion and Reason. While answering these questions you are required to choose any one of the following four responses.

A . If both Assertion and Reason are true and the Reason is correct explanation of the Assertion.

B. If both Assertion and Reason are true but the Reason is not a correct explanation of the Assertion.

C. If Assertion is true but the Reason is false.

D. If both Assertion and Reason are false

****

1. Assertion: In four o'clock plant or Snap dragon plant, a cross between homozygous white flowered individual and a homozygous red flowered one, produces pink flowered plants.

Reason: In these plants, the flower colour is determined by three alleles.

2. Assertion: It is not possible for human parents heterozygous for skin colour to have children darker or lighter than themselves.

Reason: Human skin colour is controlled by a single pair of alleles.

3. **Assertion:** The person heterozygous for sickle-cell trait produces both normal (Hb^A) and abnormal haemoglobin (Hb^s)

Reason: The normal allele and sickle allele are codominant.

4. Assertion: The defect called erythroblastosis foetalis results if Rh-negative mother carries a second Rh-positive foetus.

Reason: Replacement of the infant's blood immediately after birth often cures the defect.

5. **Assertion:** Haemophilia shows criss-cross inheritance

Reason: The gene that causes haemophilia is recessive and lies in the sex (X) chromosome.

ANSWERS:

1. C 2. D 3. A 4. B 5. A

CLASS: XII

SUBJECT: BIOLOGY

CHAPTER NUMBER: 6

CHAPTER: MOLECULAR BASIS OF INHERITANCE

ASSERTION AND REASONING TYPE OF QUESTIONS WITH ANSWERS

These questions consist of two statements each, printed as Assertion and Reason. While answering these questions you are required to choose any one of the following four responses.

A . If both Assertion and Reason are true and the Reason is correct explanation of the Assertion.

B. If both Assertion and Reason are true but the Reason is not a correct explanation of the Assertion.

C. If Assertion is true but the Reason is false.

D. If both Assertion and Reason are false

****

1. **Assertion:** Only the sense strand of DNA is copied into mRNA.

Reason: The antisense strand plays a role in replication.

2. **Assertion:** Sickle-cell haemoglobin has a valine in place of glutamic acid at position 6 in the β polypeptide chain.

Reason: Sickle-cell anaemia is expressed only in homozygous recessive state.

3. **Assertion:** DNA fingerprinting is very well known for its application in paternity testing, in case of disputes.

Reason: It employs the principle of polymorphism in DNA sequences as the polymorphisms are inheritable from parents to children.

4. **Assertion:** The genetic code is degenerate.

Reason: Most amino acids are coded by more than one codon.

5. **Assertion:** Human Genome Project was a mega project launched to find out the complete DNA sequence of human genome.

Reason: It was possible only with the help of genetic engineering techniques to isolate and clone any piece of DNA and fast techniques for determining DNA sequences.

ANSWERS:

1. B 2. B 3. A 4. A 5. A

CLASS: XII

SUBJECT: BIOLOGY

CHAPTER NUMBER: 7

CHAPTER: EVOLUTION

ASSERTION AND REASONING TYPE OF QUESTIONS WITH ANSWERS

These questions consist of two statements each, printed as Assertion and Reason. While answering these questions you are required to choose any one of the following four responses.

A . If both Assertion and Reason are true and the Reason is correct explanation of the Assertion.

B. If both Assertion and Reason are true but the Reason is not a correct explanation of the Assertion.

C. If Assertion is true but the Reason is false.

D. If both Assertion and Reason are false

****

1. **Assertion:** Cell like aggregates of complex organic compounds are called coacervates.

   Reason: Coacervates had power to grow and divide.

2. **Assertion:** Insecticides and antibiotics remain effective for an unlimited period.

   Reason: Organisms never loose sensitivity to insecticides and antibiotics.

3. **Assertion:** Acquired characters are non-inheritable.

   Reason: Acquired characters generally affect the somatic cells and it is the germ cells that are passed on to the next generation.

4. **Assertion:** Human and great apes have a common ancestry.

   Reason: Man and chimpanzee have similar banding pattern of chromosomes number 3 and 6.

5. **Assertion:** Man has descended from Monkeys.

   Reason: Monkeys resemble humans more than apes do.

ANSWERS:

1. B 2. D 3. A 4. A 5. D

CLASS: XII

SUBJECT: BIOLOGY

CHAPTER NUMBER: 8

CHAPTER: HUMAN HEALTH AND DISEASE

ASSERTION AND REASONING TYPE OF QUESTIONS WITH ANSWERS

These questions consist of two statements each, printed as Assertion and Reason. While answering these questions you are required to choose any one of the following four responses.

A . If both Assertion and Reason are true and the Reason is correct explanation of the Assertion.

B. If both Assertion and Reason are true but the Reason is not a correct explanation of the Assertion.

C. If Assertion is true but the Reason is false.

D. If both Assertion and Reason are false

****

1. **Assertion:** Morphine is the most effective analgesic.

   Reason: It binds to the postsynaptic membrane and blocks the binding of neurotransmitters.

2. **Assertion:** Smoking causes cancer of tongue, lips and lungs.

   Reason: Smoking mutates the critical gene P53 which protects the body from cancer.

3. **Assertion:** Phagocytes capture and digest microbes.

   Reason: Killer T-cells approach the body cells covered with viruses and destroy such cells before viruses can enter and multiply to spread infection.

4. **Assertion:** It is beneficial to feed the new born infants on mother's milk.

   Reason: It develops deep affection between the offspring and mother.

5. **Assertion:** Interferons help eliminate viral infections.

   Reason: Interferons released by infected cells, reach the nearby uninfected cells and make them resistant to viral infection.

ANSWERS:

1. A 2. A 3. B 4. B 5. A

CLASS: XII

SUBJECT: BIOLOGY

CHAPTER NUMBER:9

CHAPTER: STRATEGIES FOR ENHANCEMENT IN FOOD PRODUCTION

ASSERTION AND REASONING TYPE OF QUESTIONS WITH ANSWERS

These questions consist of two statements each, printed as Assertion and Reason. While answering these questions you are required to choose any one of the following four responses.

A . If both Assertion and Reason are true and the Reason is correct explanation of the Assertion.

B. If both Assertion and Reason are true but the Reason is not a correct explanation of the Assertion.

C. If Assertion is true but the Reason is false.

D. If both Assertion and Reason are false

****

1. Assertion: Hybrid seeds must be produced every year to obtain maximum advantage of heterosis.

Reason: Heterosis is lost by inbreeding.

2. Assertion: Virus-free plants can be produced from virus infected plants by means of meristematic tissue culture.

Reason: The virus fails to grow during the growth of host tissue in the artificial medium.

3. Assertion: A gene bank should not be regarded as a plant museum.

Reason: The germplasms stored in the gene bank are actively utilised by breeders to develop novel varieties.

4. Assertion: Protoplast fusion is one of the major advantages of tissue culture.

Reason: The naked protoplasts of two different plants fuse to form a hybrid.

5. Assertion: Soma clonal variations occur in tissue culture processes.

Reason: Variations cannot occur in nature.

ANSWERS:

1. A 2.C. 3.A 4.B 5.C

CLASS: XII

SUBJECT: BIOLOGY

CHAPTER NUMBER:10

CHAPTER: MICROBES IN HUMAN WELFARE

ASSERTION AND REASONING TYPE OF QUESTIONS WITH ANSWERS

These questions consist of two statements each, printed as Assertion and Reason. While answering these questions you are required to choose any one of the following four responses.

A . If both Assertion and Reason are true and the Reason is correct explanation of the Assertion.

B. If both Assertion and Reason are true but the Reason is not a correct explanation of the Assertion.

C. If Assertion is true but the Reason is false.

D. If both Assertion and Reason are false

****

1. Assertion: Chemical pesticides are more hazardous as compared to biopesticides.

Reason: Chemical pesticides are mostly non-specific, expensive, hazardous and pollute the atmosphere.

2. Assertion: Yeasts should not be used in brewing and baking industries.

Reason: They produce several harmful products during brewing and baking.

3. Assertion: Most of the orchids are dependent on fungi in seeding stage.

Reason: The fungal mycelium penetrates the cortical cells of the orchid roots.

4. Assertion: Industrial fermentations are money making ventures.

Reason: They require a through research to discover high yields of produce at least possible expenses.

5. Assertion: Leguminous plants are best preferred for rotation of crops.

Reason: They have root nodule which have nitrogen fixing bacteria Clostridium.

ANSWERS:

1. A 2. D 3. B 4. B 5. C

CLASS: XII

SUBJECT: BIOLOGY

CHAPTER NUMBER:11

CHAPTER: BIOTECHNOLOGY:PRINCIPLES AND PROCESSES

ASSERTION AND REASONING TYPE OF QUESTIONS WITH ANSWERS

These questions consist of two statements each, printed as Assertion and Reason. While answering these questions you are required to choose any one of the following four responses.

A . If both Assertion and Reason are true and the Reason is correct explanation of the Assertion.

B. If both Assertion and Reason are true but the Reason is not a correct explanation of the Assertion.

C. If Assertion is true but the Reason is false.

D. If both Assertion and Reason are false

****

1. **Assertion:** Exonucleases remove nucleotides from the ends of DNA.

Reason: Endonucleases make cuts at specific positions within the DNA.

2. **Assertion:** The cut pieces of DNA are linked with plasmid DNA.

Reason: Plasmid DNA fails to act as Vectors.

3. **Assertion:** Recombinant DNA technology has become successful because of the presence of restriction endonucleases in eukaryotic cells.

Reason: Restriction endonucleases cut the DNA molecule to form blunt ends.

4. **Assertion:** Genetic engineering overcomes the drawbacks of traditional hybridisation.

Reason: Genetic engineering involves creation of a recombinant DNA and introduce the desirable genes into target organisms.

5. **Assertion:** Asexual reproduction preserves the genetic information.

Reason: Sexual reproduction permits variation.

ANSWERS:

1. B **2. C** **3. D** **4. A** **5. B**

CLASS: XII

SUBJECT: BIOLOGY

CHAPTER NUMBER:12

CHAPTER: BIOTECHNOLOGY AND ITS APPLICATIONS

ASSERTION AND REASONING TYPE OF QUESTIONS WITH ANSWERS

These questions consist of two statements each, printed as Assertion and Reason. While answering these questions you are required to choose any one of the following four responses.

A . If both Assertion and Reason are true and the Reason is correct explanation of the Assertion.

B. If both Assertion and Reason are true but the Reason is not a correct explanation of the Assertion.

C. If Assertion is true but the Reason is false.

D. If both Assertion and Reason are false

****

1. Assertion: In recombinant DNA technology, human genes are often transferred into bacteria(prokaryotic) or yeast (eukaryotic)

Reason: Both bacteria and yeast multiply very fast to form huge population which express the desired gene.

2. Assertion: *Agrobacterium tumefaciens* is popular in genetic engineering because this bacterium is associated with the roots of all cereal and pulse crops.

Reason: A gene incorporated in the bacterial chromosomal genome gets automatically transferred to the crop with which bacterium is associated.

3. Assertion: Monoclonal antibodies are ideal for diagnosis of diseases caused by closely related pathogens.

Reason: Monoclonal antibodies are far more specific and reproducible than the antibodies produced by conventional techniques.

4. Assertion: Biopiracy is being resorted to by the countries of North.

Reason: Countries of South are incapable of Biopiracy.

5. Assertion: Cholesterol is one of the most important steroids in animals and human beings.

Reason: It is present as a component of animal cell walls and acts as an agent of cell wall extension.

ANSWERS:

1. A **2. D** **3. A** **4. C** **5. C**

CLASS: XII

SUBJECT: BIOLOGY

CHAPTER NUMBER:13

CHAPTER: ORGANISMS AND POPULATIONS

ASSERTION AND REASONING TYPE OF QUESTIONS WITH ANSWERS

These questions consist of two statements each, printed as Assertion and Reason. While answering these questions you are required to choose any one of the following four responses.

A . If both Assertion and Reason are true and the Reason is correct explanation of the Assertion.

B. If both Assertion and Reason are true but the Reason is not a correct explanation of the Assertion.

C. If Assertion is true but the Reason is false.

D. If both Assertion and Reason are false

****

1. **Assertion:** Chamaeleon can change its colour.

   Reason: It is a fashionable animal.

2. **Assertion:** Many hydrophytes lack roots.

   Reason: Roots are used for balancing.

3. **Assertion:** Scavenging improves the environment.

   Reason: A scavenger disposes of dead organic matter.

4. **Assertion:** Pathogenic bacteria do not multiply well in soil.

   Reason: Antibiotic producing bacteria and fungi are common in the soil.

5. **Assertion:** Parasites produce fewer young ones.

   Reason: They have an easy life in their hosts.

ANSWERS:

1. C 2. B 3. A 4. A 5. D

CLASS: XII

SUBJECT: BIOLOGY

CHAPTER NUMBER:14

CHAPTER: ECOSYSTEM

ASSERTION AND REASONING TYPE OF QUESTIONS WITH ANSWERS

These questions consist of two statements each, printed as Assertion and Reason. While answering these questions you are required to choose any one of the following four responses.

A . If both Assertion and Reason are true and the Reason is correct explanation of the Assertion.

B. If both Assertion and Reason are true but the Reason is not a correct explanation of the Assertion.

C. If Assertion is true but the Reason is false.

D. If both Assertion and Reason are false

****

1. **Assertion:** Oceans acts as the global 'sink' for CO_2.

   Reason: Human activities are increasing CO_2 concentration in the air.

2. **Assertion:** Lichens are a pioneer community in the temperate region.

   Reason: Blue green algae grow as the pioneer community in the tropical region.

3. **Assertion:** Ecological succession can turn a lake into a dryland forest with time.

   Reason: A bare rock can become a forest through ecological succession.

4. **Assertion:** Life occurs up to 15,000 metres on the mountains.

   Reason: Beyond 15,000 metres, temperature is too low , CO_2 and O_2 are also too little.

5. **Assertion:** Ozone layer is present in the stratosphere.

   Reason: Hydrosphere covers about 70% of the earth's surface.

ANSWERS:

1. B 2. B 3. B 4. A 5. B

CLASS: XII

SUBJECT: BIOLOGY

CHAPTER NUMBER:15

CHAPTER: ENVIRONMENTAL ISSUES

ASSERTION AND REASONING TYPE OF QUESTIONS WITH ANSWERS

These questions consist of two statements each, printed as Assertion and Reason. While answering these questions you are required to choose any one of the following four responses.

A . If both Assertion and Reason are true and the Reason is correct explanation of the Assertion.

B. If both Assertion and Reason are true but the Reason is not a correct explanation of the Assertion.

C. If Assertion is true but the Reason is false.

D. If both Assertion and Reason are false

****

1. **Assertion:** Radioactive pollution may affect the future generation.

Reason: It causes mutations and mutant genes may affect the progeny.

2. **Assertion:** Water logged soil is physiologically dry.

Reason: Water dislodges air from the soil so that roots cannot breathe.

3. **Assertion:** Population density of indicator organisms shows degree of water pollution.

Reason: Low BOD indicates less organic waste in water.

4. **Assertion:** A suspended particulate matter (SPM) is an important pollutant released by diesel vehicles.

Reason: Catalytic converters greatly reduce pollution caused by automobiles.

5. **Assertion:** The concentration of methane in the atmosphere has more than doubled in the last 250 years.

Reason: Wetlands and rice fields are the major sources of methane.

ANSWERS:

1. A **2. A** **3. B** **4. B** **5. A**

CLASS: XII

SUBJECT: BIOLOGY

CHAPTER NUMBER:16

CHAPTER: BIODIVERSITY AND CONSERVATION

ASSERTION AND REASONING TYPE OF QUESTIONS WITH ANSWERS

These questions consist of two statements each, printed as Assertion and Reason. While answering these questions you are required to choose any one of the following four responses.

A . If both Assertion and Reason are true and the Reason is correct explanation of the Assertion.

B. If both Assertion and Reason are true but the Reason is not a correct explanation of the Assertion.

C. If Assertion is true but the Reason is false.

D. If both Assertion and Reason are false

****

1. **Assertion:** Cleanliness resorted to by man is harmful for the wild life.

Reason: It deprives the scavengers (Vultures, Kites) of food and also destroys their habitats.

2. **Assertion:** Threatened species should be given preference over others in the conservation programmes.

Reason: All the life supporting systems (air, water and land) should be conserved.

3. **Assertion:** Tiger population of India has increased to 4000 from 1827 in 1972.

Reason: Tiger is a prolific breeder.

4. **Assertion:** Using dung as fuel greatly reduces grain production.

Reason: Dung used as manure increases crop yield.

5. **Assertion:** Forests reduce atmospheric pollution.

Reason: Forests increase rainfall.

ANSWERS:

1. A **2. B** **3. C** **4. A** **5. B**